

Deployment Documentation — ExoHabitAI

1. Project Title

ExoLifeAI — Exoplanet Habitability Prediction System

A Machine Learning–powered web application designed to predict the habitability of exoplanets based on planetary and stellar parameters.

3. Deployment Platform

Component	Technology Used
Cloud Platform	Render
Backend Framework	Flask
Frontend	HTML, CSS, JavaScript
3D Visualization	Three.js
Animations	GSAP
ML Models	Random Forest, XGBoost
WSGI Server	Gunicorn
Version Control	Git & GitHub

5. GitHub Repository

Repository Link:

<https://github.com/GKSJ19/ExoHabitAI/tree/SanjanaPandurangMalkhede>

Repository contains:

- Backend API code
 - Frontend UI files
 - Trained ML models
 - Requirements file
 - Deployment configuration
-

6. Live Deployment URL

Deployed Application:

<https://exohabitai-m1ku.onrender.com/>

The application is publicly accessible and performs live habitability predictions.

7. Deployment Steps

Step 1 — Repository Preparation

- Project code organized into backend and frontend modules.
 - Model files serialized using Joblib/Pickle.
 - requirements.txt created.
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Step 2 — Repository Fork

- Organization repository forked to personal GitHub.
 - Ensured deployment permissions.
-

Step 3 — Render Web Service Creation

- GitHub account connected to Render.
 - New Web Service created.
 - Repository and branch selected.
-

Step 4 — Build Configuration

Build Command:

```
pip install -r requirements.txt
```

Installs all Python dependencies required by the Flask backend and ML models.

Step 5 — Start Command

```
gunicorn backend.app:app
```

Runs the Flask application using Gunicorn production server.

Step 6 — Static File Configuration

Frontend files served using Flask static routing:

```
app = Flask(
    __name__,
    static_folder="./frontend",
    static_url_path="/static"
)
```

Ensures proper delivery of CSS and JavaScript assets.

Step 7 — Deployment Execution

- Build triggered on Render.
 - Dependencies installed.
 - Models loaded.
 - Gunicorn server started.
 - Live URL generated.
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8. API Endpoints Implemented

8.1 Habitability Prediction API

Endpoint:

POST /predict

Function:

- Accepts planetary & stellar parameters.
- Runs ML inference.
- Returns classification and probability.

Response Format:

```
{
  "status": "success",
  "prediction": {
    "habitability": "Habitable",
    "score": 0.92
  }
}
```

```
}  
}
```

9. API Testing & Validation

APIs were tested using:

- Frontend form submission
- Browser network calls
- JSON response validation

All endpoints returned expected outputs without errors.

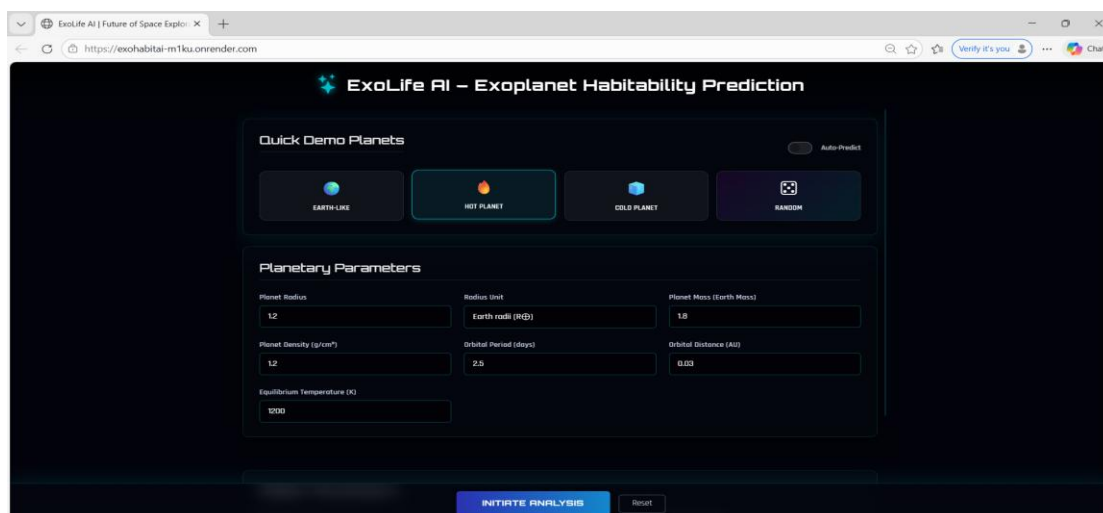
10. Model Deployment Verification

During deployment logs:

- ML models loaded successfully.
 - No runtime errors observed.
 - Predictions generated in real time.
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11. Screenshot Evidence

Screenshots attached include:



ExoLife AI – Exoplanet Habitability Prediction

Planetary Parameters

Planet Radius 1.2	Radius Unit Earth radii (R _⊕)	Planet Mass (Earth Mass) 1.8
Planet Density (g/cm³) 1.2	Orbital Period (days) 2.5	Orbital Distance (AU) 0.03
Equilibrium Temperature (K) 1200		

Stellar Parameters

Star Effective Temperature (K) 6200	Star Luminosity 0.1	Star Metallicity -0.05
Star Type ☒ F ☐ G ☐ K ☐ M		

INITIATE ANALYSIS **Reset**

ExoLife AI – Exoplanet Habitability Prediction

Quick Demo Planets Auto-Predict

☒ EARTH-LIKE
 ☐ HOT PLANET
 ☐ COLD PLANET
 ☐ RANDOM

ANALYSIS REPORT

NOT HABITABLE

HABITABILITY SCORE
12.00%

← New Prediction

INITIATE ANALYSIS **Reset**

12. Challenges Faced

Issue	Resolution
Static files not loading	Configured static_url_path
API fetch failure	Updated production API URL
Branch deployment access	Forked repository
Large dataset warnings	Ignored for deployment

13. Deployment Outcome

- Application deployed successfully.
 - Backend APIs functional.
 - ML predictions operational.
 - UI rendered correctly.
 - Public URL accessible.
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14. Conclusion

The ExoLifeAI system was successfully deployed on the Render cloud platform. The integration of Machine Learning models with a Flask backend and an interactive 3D frontend interface enables real-time exoplanet habitability prediction, demonstrating full-stack AI deployment capability.